

HUZAIF KHAN

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Full-Stack Software Engineer with 5+ years building production software across enterprise mobile platforms, cloud-native backends, and AI/ML systems. Proven track record shipping high-impact features at scale, publishing peer-reviewed ML research, and leading cross-functional engineering teams from design through deployment.

SKILLS

Languages: Python, Java, C#, C, SQL, JavaScript, TypeScript, Scala

Frameworks & Libraries: React.js, Next.js, Node.js, Angular, Django, FastAPI, Express.js, Xamarin, React Native, PyTorch, Pydantic Cloud & DevOps: AWS (Lambda, API Gateway, DynamoDB, Cognito, S3, CloudWatch, IAM), Azure SQL, Docker, Kubernetes, GitHub Actions, CI/CD, Git, Postman, UNIX/Linux

Data & Messaging: PostgreSQL, MySQL, Redis, DynamoDB, Azure SQL Server, REST APIs, JWT, JSON, WebSockets

AI/GenAI Tooling: LLMs, NLP, Federated Learning, Homomorphic Encryption, BioBERT, OpenAI GPT, TF-IDF, SVM, PyTorch, LangChain, RAG, Fine-Tuning

WORK EXPERIENCE

CALIFORNIA STATE UNIVERSITY, DOMINGUEZ HILLS Jan 2024 - Present · Los Angeles, CA

Research Assistant - VR/XR with AI & ML

- Build multiplayer VR classroom applications in **Unity 6 / C# (URP)** for Meta Quest, enabling students to interact, collaborate, and study together in shared, immersive 3D learning environments.
- Engineered networked multi-user sessions with **Unity Netcode for GameObjects (NGO)** and **Unity Relay**, implementing state synchronization, connection management, and client-server authority for low-latency real-time interaction.
- Developing integration of **AI teaching agents** (LLM-driven) into live VR classrooms, wiring real-time inference into the game loop to deliver interactive, on-demand instruction to students.

CALIFORNIA STATE UNIVERSITY, DOMINGUEZ HILLS Mar 2024 - May 2026 · Los Angeles, CA

Research Assistant - Security, Edge AI & ML

- Published six peer-reviewed works including at CSNET 2024 and ICNC 2025, presented a poster at IEEE CCNC 2025, and peer-reviewed two IEEE AIoT 2026 submissions in computer vision and robotics.
- Engineered a federated learning pipeline in Python integrating homomorphic encryption with optimized K-means clustering, reducing false positive rates by 30%+ and strengthening ML model integrity across distributed data sources.

CAPGEMINI TECHNOLOGY SERVICES Jul 2021 - Jul 2023 · Bangalore, India

Senior Software Analyst

- Led end-to-end redevelopment of a mission-critical enterprise mobile application in Xamarin and C#, delivering 50%+ improvements in stability, performance, and user experience for a high-profile UK client serving thousands of daily active users.
- Designed and deployed Azure SQL Server triggers and scheduled diagnostic jobs to automate time-log validation and anomaly detection, eliminating false alarms and ensuring 99%+ data reliability across production pipelines.
- Optimized backend query performance through strategic composite indexing, query refactoring, and client-side pagination, reducing average API response latency by 40% under peak concurrent load.

CAPGEMINI TECHNOLOGY SERVICES Jan 2021 - Apr 2021 · Bangalore, India

Full-Stack Developer Intern

- Led a team of 4 engineers to design and ship a cross-platform e-commerce application in React Native and Node.js, delivering a fully functional product end-to-end within a 4-month timeline.
- Engineered a custom payment API with AES encryption, JWT token-based authentication, and structured transaction logging, ensuring enterprise-grade security and auditability across all user sessions.
- Reduced average page load time by 2 seconds by implementing lazy loading for product images, code splitting, and deferring non-critical scripts, directly improving Core Web Vitals and user retention metrics.

THINGSCLLOUD TECHNOLOGIES PVT LTD Jul 2019 - Aug 2019 · Bangalore, India

Frontend Developer Intern

- Developed reusable Angular 7/TypeScript components for a production IoT platform, improving UI consistency and cutting rendering overhead under high-frequency data updates via DOM update batching and OnPush change detection.

PROJECTS

SenseCLLM: Edge-Resident PII Detection 2026

Python, Ollama, Multi-Agent Systems, Transformers, PyTorch, Cryptography

- Engineered a three-agent, edge-resident PII detection framework running LLMs fully on-device via Ollama, integrating a Transformers/PyTorch inference pipeline with a pycryptodome-based encryption layer to enable privacy-preserving processing of clinical text without cloud exposure, accepted at IEEE AIoT 2026.

Medical Text Classification 2025

Python, BioBERT, TF-IDF, SVM, PyTorch

- Built a multi-stage NLP classification pipeline for 32 cancer types, combining TF-IDF + SVM as a reproducible baseline with fine-tuned BioBERT, applying domain-specific preprocessing and a custom clinical masking dictionary to prevent label leakage and ensure inference integrity at scale.

EDUCATION

CALIFORNIA STATE UNIVERSITY, DOMINGUEZ HILLS May 2025 · Los Angeles, CA

Master of Science, Computer Science - GPA: 3.95/4.0